

Continuous and Synoptic Assessment of Indian Inland Waters for Harmful Algae Blooms

Chintan B. Maniyar^{a,b}, Abhishek Kumar^{b,c,*}, Deepak R. Mishra^b

^a Photogrammetry and Remote Sensing Department, Indian Institute of Remote Sensing (IIRS), ISRO, India

^b Department of Geography, University of Georgia, GA, USA

^c Department of Environmental Conservation, University of Massachusetts Amherst, MA, USA

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ABSTRACT

Cyanobacterial Harmful Algal Blooms (CyanoHABs) are progressively becoming a major water quality, socio-economic, and health hazard worldwide. In India, there are frequent episodes of severe CyanoHABs, which are left untreated due to a lack of awareness and monitoring infrastructure, affecting the economy of the country gravely. In this study, for the first time, we present a country-wide analysis of CyanoHABs in India by developing a novel interactive cloud-based dashboard called “CyanoKhoj” in Google Earth Engine (GEE) which uses Sentinel-3 Ocean and Land Colour Instrument (OLCI) remotely sensed datasets. The main goal of this study was to showcase the utility of CyanoKhoj for rapid monitoring and discuss the widespread CyanoHABs problems across India. We demonstrate the utility of CyanoKhoj by including select case studies of lakes and reservoirs geographically spread across five states: Bargi and Gandhisagar Dams in Madhya Pradesh, Hirakud Reservoir in Odisha, Ukai Dam in Gujarat, Linganamakki Reservoir in Karnataka, and Pulicat Lake in Tamil Nadu. These sites were studied from September to November 2018 using CyanoKhoj, which is capable of near-real-time monitoring and country-wide assessment of CyanoHABs. We used CyanoKhoj to prepare spatiotemporal maps of Chlorophyll-a (Chl-a) content and Cyanobacterial Cell Density (CCD) to study the local spread of the CyanoHABs and their phenology in these waterbodies. A first-ever all-India CCD map is also presented for the year 2018, which highlights the spatial spread of CyanoHABs throughout the country (32 large waterbodies across India with severe bloom: CCD > 2,500,000). Results indicate that CyanoHABs are most prevalent in nutrient-rich waterbodies prone to industrial and other nutrient-rich discharges. A clear temporal evolution of the blooms showed that they are dominant during the post-monsoon season (September–October) when the nutrient concentrations in the waterbodies are at their peak, and they begin to decline towards winter (November–December). CyanoKhoj is an open-source tool that can have a significant broader impact in mapping CyanoHABs not only throughout cyanobacteria data-scarce India, but on a global level using archived and future Sentinel-3A/B OLCI data.

1. Introduction

Bloom forming cyanobacteria or blue-green algae, when left untreated can be a major threat to inland waters and estuarine environments. Cyanobacterial Harmful Algal Blooms (CyanoHABs) adversely affect aquatic ecosystem services and public lives (Mishra et al., 2020a) by causing fish kills, animal deaths, human health hazards and, disruption in food and water supply (Ibelings et al., 2014; R.P. Stumpf et al., 2016). The most recent shocking example of CyanoHABs impact came from Botswana where nearly 400 elephants died from ingesting cyanobacteria from water (BBC News, 2020; van Aarde et al., 2021;

Veerman et al., 2021). The prevalence of CyanoHAB's in a waterbody also hampers recreational activities and negatively affects economic activities. Climate change and nutrient pollution are the big drivers of the global proliferation of CyanoHABs, as Paerl and Huisman (2008) put it, “blooms like it hot”. Rising global temperatures, changing precipitation patterns, and drought frequency and severity, and combinedly, all help trigger CyanoHABs (Paerl and Otten, 2013). Inherently cold polar regions such as the Himalayas and Antarctica are also becoming susceptible to such toxic harmful algal blooms (Gray et al., 2020; Inaotombi and Sarma, 2020; Republic World, 2020). In 2019 alone, which happened to be the hottest year on record, there have been numerous

* Corresponding Author.

E-mail address: akumar1@umass.edu (A. Kumar).

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reports of CyanoHABs in various Indian waterbodies and reservoirs (Inaotombi and Sarma, 2020; Mohan et al., 2020; Narayana et al., 2020). Bargi and Gandhisagar Dams in Madhya Pradesh, Ukai Dam in Gujarat, Stanley Reservoir in Tamil Nadu, Chilika Lagoon in Odisha are only a handful of examples of affected Indian inland waterbodies apart from the coastal waters in the east and west coasts of India (Mishra et al., 2020a; J.G. Ray et al., 2020). A decade-old study in 2012 which reviewed CyanoHABs occurrences in India indicated an increase as steep as 15 percent in the past 12 years (D'Silva et al., 2012; The Hindu, 2010). It is safe to assume that there has been an increase in the rate of CyanoHABs occurrences in India since 2012 (Bharati and Singh, 2020; Jasser et al., 2020; J.G. Ray et al., 2020; Singh and Babele, 2020).

A vast majority of the Indian population is deeply connected to their surrounding waterbodies either for their livelihoods such as fishing and agriculture or day-to-day lives such as bathing, cooking, and spiritual rituals. With their well-established negative health impact (O'Keefe and Juliette, 2019), these blooms have the potential to be a public health disaster during a bloom-affected year. Frequent exposure to CyanoHABs can result in seafood poisoning, nausea, breathlessness, diseases triggered by neurotoxins and hepatotoxins such as Dementia, Alzheimer's and other neurological impairments, gastrointestinal and respiratory problems, and adverse impact on the liver and kidney functioning (Carmichael et al., 1985; R.P. Stumpf et al., 2016; Zhang et al., 2015). Over the years, numerous people have been hospitalized and even died in Tamil Nadu, Kerala, Maharashtra, and coastal regions of India owing to the effects of CyanoHABs (Hindu, 2010; Padmakumar et al., 2012). Many more may have gone unreported. These CyanoHABs are more likely to disproportionately impact the poorest sections of the society because of their direct dependence on water resources and their negative impact on ecosystem dynamics. Any negative health impact due to their exposure to CyanoHABs will exacerbate the vicious cycle of illness-induced poverty as it is now well recognized that poverty and diseases are closely linked, not only because poor people are more likely to get sick, but also because increased sickness exacerbates poverty (Ngonghala et al., 2014). Without government intervention in the form of stricter environmental regulations, we can expect to see such blooms in the marine and freshwater bodies of India more frequently. Despite the animal, environmental, and human health risks, there is no established rapid monitoring or forecasting program in India to evaluate the initiation, spread, and spatial distribution of CyanoHABs. To our knowledge, there is also no proactive nationwide system to assist people in making safe water usage decisions. This implies more endangerment to human lives, the aquaculture food web, and the ecosystem services of the country. India's ecosystem services, while amounting to less than 10% of national gross domestic product (GDP), can amount to over 50% of the income of India's rural poor (Sukhdev, 2009). Therefore, it is imperative to have a robust infrastructure for near-real-time detection and monitoring of Indian waterbodies for CyanoHABs in the form of an early warning system.

It is often difficult to differentiate a CyanoHAB from regular phytoplankton blooms because both can be similar in visual appearance. Some studies have detected CyanoHABs using chlorophyll-a (Chl-a) concentration as a proxy via different remote sensing indices and bio-optical models (Clark et al., 2017; Kumar et al., 2020; Mishra et al., 2018; Page et al., 2018; Pettersson et al., 2013; Sayers et al., 2019; Schaeffer et al., 2018; Wynne et al., 2010). These models could overestimate cyanobacteria content in the waterbodies since Chl-a pigment is present in both phytoplankton and cyanobacteria. Therefore, studies focused on CyanoHABs detection often use a characteristic pigment called Phycocyanin (PC) whose optical property is different than Chl-a (Mishra et al., 2013; Mishra and Mishra, 2014; Ogashawara et al., 2013; Simis et al., 2005; Beck et al., 2017; Yan et al., 2018). Some studies have also used cyanobacteria cell density (CCD) to estimate cyanobacteria content in the water (Lunetta et al., 2015; Page et al., 2018; Wynne et al., 2008; (Weber et al., 2020)). Traditional on-field and in-situ measurement methods are labor-intensive and expensive, and given the

spatio-temporal variability of CyanoHABs (Sayers et al., 2019), it makes near-real-time monitoring and assessment difficult. Satellite-based methods are the only viable options for rapid and frequent detection and monitoring of the majority of the water bodies across India. In addition, gathering a variety of ancillary data from different sources and combinedly processing them with remote sensing data is an effective means to create an early information system for CyanoHABs (Mishra et al., 2020a). Remote sensing derived big data analytics is an efficient technology in terms of time and efforts for such an early information objective (Chi et al., 2016). CyanoHABs, too can be monitored in such a timely fashion for early detection and warning using data analytics and cloud computing. In this study, we have used Google Earth Engine (GEE) as the cloud-based computation platform (Gorelick et al., 2017) to develop "CyanoKhoj" (meaning "search for cyanobacteria" in Hindi), an early warning monitoring system for CyanoHABs in India using the European Space Agency's (ESA) Sentinel-3 Ocean and Land Color Instrument (OLCI) Level-1 data (CyanoKhoj, 2020a).

Our objective was to develop and showcase a near-real-time cloud-based tool for rapid and country-wide monitoring of CyanoHABs in Indian waters, with a motivation to make it available to the water managers for water quality related decision making. 'CyanoKhoj', for analysis and visualization of CyanoHABs allows for easy detection, mapping, and dissemination of warnings to water managers or policymakers, and concerned citizens for issuing place-based alerts. We implemented a process to estimate Rayleigh corrected reflectance (Page et al., 2018) from the Sentinel-3 OLCI data in the dashboard itself for modeling and prediction of both CCD and Chl-a to differentiate between CyanoHABs and regular phytoplankton blooms. This is the first of its kind tool developed to keep track of Indian inland and coastal waters that does not use Chl-a as a proxy and specifically targets cyanobacteria concentrations. This tool is timely and much needed for India, which is becoming increasingly prone to CyanoHABs as the country becomes more urbanized, industrialized, and polluted. Unless monitored and reported, these CyanoHABs in densely populated areas could substantially increase people's exposure to cyanotoxins creating a public health disaster.

CyanoKhoj uses CyanoHABs-specific models (Wynne et al., 2010; Lunetta et al., 2015) on Rayleigh-corrected Sentinel-3 OLCI dataset, which has a 620 nm band that allows differentiating between cyanobacteria and other common algae. PC is a dominant pigment in CyanoHABs, which absorbs the incoming light strongly around 620 nm wavelength and therefore produces a dip around this band in reflectance spectra. However, to get this subtle absorption feature in satellite data requires the removal of atmospheric noises. Numerous studies have used Rayleigh correction to remove the noise from satellite data before implementing CyanoHABs specific models (Lunetta et al., 2015; Mishra et al., 2020a; Page et al., 2018). However, CyanoKhoj is a novel platform to perform Rayleigh correction of Sentinel 3-OLCI data on the cloud. Apart from Rayleigh correction, other novel aspects of CyanoKhoj include user control to re-parameterize the CyanoHABs model based on site-specific data (as open-access code), and the ability to download the map and data for further validation and analysis. Moreover, a similar mechanism can be transferred to other satellite data as well as other water quality parameters that can be derived using remote sensing techniques.

2. Data and Methodology

Abbreviations: CyanoHABs=Cyanobacterial Harmful Algal Blooms; BRR=Bottom of Rayleigh Reflectance; CCD=Cyanobacterial Cell Density; Chl-a=Chlorophyll-a; CI=Cyanobacteria Index; CWMI=Composite Water Management Index; ESA=European Space Agency; GEE=Google Earth Engine; L=Radiance; L^{TOA} =Top of Atmosphere Radiance; MAPE=Mean Absolute Percentage Error; MERIS=Medium Resolution Imaging Spectrometer; NDCI=Normalised Difference Chlorophyll Index; NOAA=National Oceanic and Atmospheric Administration;

NRMSE=Normalised Root Mean Squared Error; OLCI=Ocean and Land Color Instrument; PC=Phycocyanin; r =Reflectance; r^{BRR} =Rayleigh Corrected Reflectance; RMSE=Root Mean Squared Error; r^{TOA} =Top of Atmosphere Reflectance; SNAP=Sentinel Application Platform; SS=Spectral Shape; TOA=Top of Atmosphere; ZLD=Zero Liquid Discharge

CyanoKhoj was developed on GEE using the ESA's Sentinel-3 OLCI data. Sentinel-3 multispectral OLCI sensor has 21 spectral bands at 300 m spatial resolution with worldwide coverage every ~2 days. Given the unique absorption characteristic of cyanobacteria at 620 nm, Sentinel-3 was chosen out of all functional non-commercial satellites owing to its dedicated band at 620 nm (Band 7). For the initial processing stage dedicated to deriving coefficients for Rayleigh correction and implementing it within CyanoKhoj, Sentinel 3-OLCI data for diverse inland lakes, reservoirs, and coastal lagoons all around India (Barghi and Gandhisagar Dams in Madhya Pradesh; Ukai Lake in Gujarat; Somasila Reservoir in Andhra Pradesh; Chilika Lake in Odisha) were collected. A diverse selection of waterbodies was made in order to account for the geographical and ecological variance while deriving the Rayleigh correction coefficients using statistical analysis. These satellite datasets were collected monthly (one cloud-free scene per month for each site) for the year 2017, to account for the effect of seasonal variance. Sentinel 3-OLCI data starts from October 2016 in GEE, but we needed a complete year data (e.g., 2017) for deriving Rayleigh correction coefficients. The whole process was divided into two main tiers – Sentinel Application Platform (SNAP) tool processing and GEE processing as shown in Fig. 1. All processing steps are described in the subsequent sections.

2.1. SNAP Processing

The Level-1 (L1) Sentinel 3-OLCI Earth Observation Full Resolution (EFR) datasets were downloaded from Copernicus Online Data Access (CODA: <https://coda.eumetsat.int/#/home>). These datasets contain top-of-atmosphere (TOA) radiances (L^{TOA}) at 21 spectral bands with center wavelengths ranging from the visible to the near-infrared (400 to 1029 nm). During pre-processing, the L1 data were used as an input file

in the SNAP toolbox to derive the bottom of Rayleigh corrected reflectance (r^{BRR}) and TOA reflectance (r^{TOA}). After pre-processing, r^{TOA} and r^{BRR} values were extracted corresponding to water pixels of various waterbodies, including inland freshwater, saline water, and brackish water sites. Finally, a linear regression analysis was carried between extracted per-pixel r^{TOA} and r^{BRR} data to derive empirical coefficients that can be implemented in GEE platform to derive r^{BRR} (GEE) from OLCI without metadata.

2.2. GEE Processing

2.2.1. TOA Reflectance

The OLCI data available in GEE are L1 TOA radiance products that lack metadata (such as solar zenith and azimuth angles, satellite viewing angle). However, for independent pre-processing of OLCI scenes, we need a metadata file. Therefore, as an alternative solution, we established a correlation between SNAP processed OLCI pixels r^{TOA} and corresponding pixels derived L^{TOA} from GEE platform (Eq. (1)).

$$r_{\lambda}^{TOA}(GEE) \approx r_{\lambda}^{TOA}(SNAP) = m_{\lambda}^L \times L_{\lambda}^{TOA}(GEE) + c_{\lambda}^L \quad (1)$$

Where; $r_{\lambda}^{TOA}(GEE)$ and $r_{\lambda}^{TOA}(SNAP)$ are TOA reflectance derived using GEE and SNAP platform respectively for a particular wavelength (λ); $L_{\lambda}^{TOA}(GEE)$ is TOA radiance for a particular wavelength (λ) derived using GEE; m_{λ}^L is correlation slope between $r_{\lambda}^{TOA}(SNAP)$ and $L_{\lambda}^{TOA}(GEE)$ and c_{λ}^L is the intercept for a particular wavelength (λ). Coefficient values for both m_{λ}^L and c_{λ}^L are provided in. Suppl. Table 1.

2.2.2. Rayleigh Corrected Reflectance

To further correct $r^{TOA}(GEE)$ for Rayleigh scattering and obtain $r^{BRR}(GEE)$, $r^{BRR}(SNAP)$ and $r^{TOA}(SNAP)$ were correlated for each band (Eq. (2)). Hence, for a particular band with wavelength (λ):

$$r_{\lambda}^{BRR}(GEE) \approx r_{\lambda}^{BRR}(SNAP) = m_{\lambda}^R \times r_{\lambda}^{TOA}(SNAP) + c_{\lambda}^R \quad (2)$$

Where; $r_{\lambda}^{BRR}(GEE)$ is Rayleigh corrected reflectance derived within GEE interface; $r_{\lambda}^{BRR}(SNAP)$ is Rayleigh corrected reflectance derived within SNAP; $r_{\lambda}^{TOA}(SNAP)$ is TOA reflectance derived within SNAP

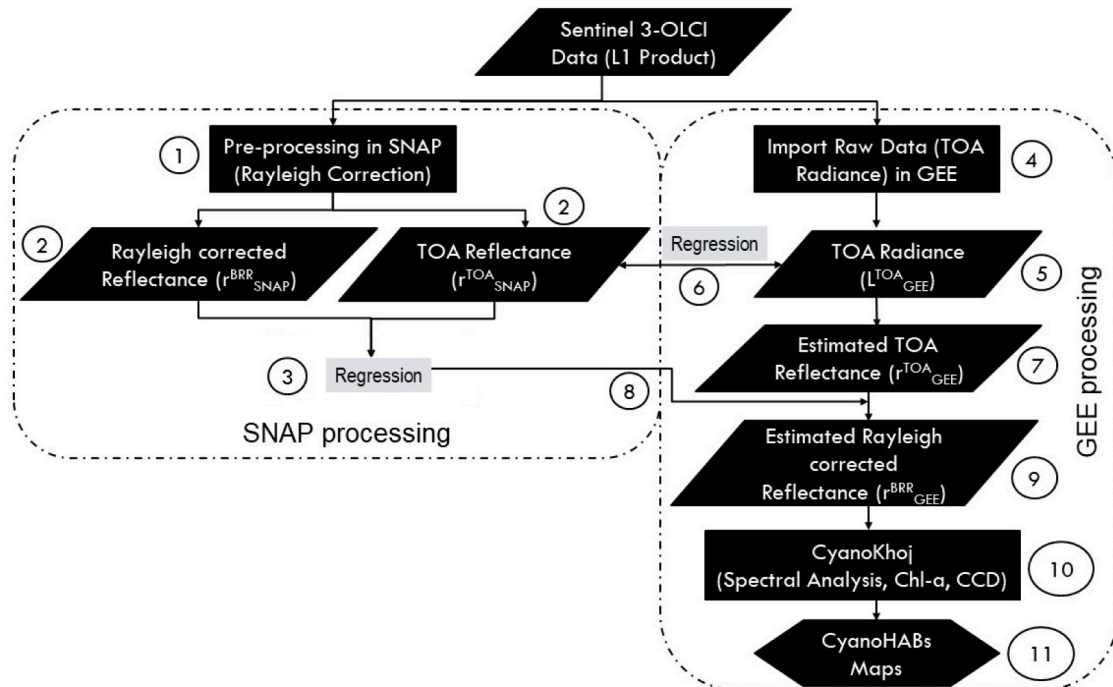


Fig. 1. Overall methodology including data, processing platforms (GEE: Google Earth Engine and SNAP Toolbox), and various processing levels (numbered) to derive final Cyanobacterial Harmful Algal Blooms (CyanoHABs) maps. The abbreviations used in the flowchart are as follow: r^{BRR} = Bottom of Rayleigh Reflectance; TOA = Top of Atmosphere; r = Reflectance; L = radiance; Chl-a = Chlorophyll-a; CCD = Cyanobacteria Cell Density.

platform for a particular wavelength (λ); m_{λ}^R is correlation slope between $r_{\lambda}^{TOA}(\text{SNAP})$ and $r_{\lambda}^{BRR}(\text{SNAP})$ and c_{λ}^R is the intercept for a particular wavelength (λ). Coefficient values for both m_{λ}^R and c_{λ}^R are provided in Suppl. Table 2.

2.3. Errors and Uncertainties

To evaluate the performance of Rayleigh correction within GEE interface, Sentinel 3-OLCI images corresponding to a peak bloom period (September–October 2017) in India were used (Suppl. Table 3). The band-wise error in the conversion of TOA radiance (L_{λ}^{TOA}) to TOA reflectance (r_{λ}^{TOA}) as well as conversion of TOA reflectance to Rayleigh corrected reflectance (r_{λ}^{BRR}) within GEE interface was estimated using percentage Normalized Root Mean Squared Error (%NRMSE) metric Eqs. (3)–(6). A total of 635 randomly distributed pixels across all satellite images of all five sites were used to perform error estimation (Suppl. Table 3). For error estimation in the conversion of L_{λ}^{TOA} to r_{λ}^{TOA} within GEE, $r_{\lambda}^{TOA}(\text{SNAP})$ was considered as actual observation and $r_{\lambda}^{TOA}(\text{GEE})$ was considered as an estimated observation for each pixel (as shown in Eq. (3)). SNAP software was specifically developed for Sentinel (1,2,3,5) satellite platforms to implement validated algorithms and was used in numerous studies for atmospheric correction (Page et al., 2018; Kyrilyuk and Kratzer, 2019; Kumar et al., 2020; Mishra et al., 2020a). Therefore, we considered reflectance derived from SNAP platform as reference in the absence of actual field measurement.

$$RMSE_{TOA} = \sqrt{\frac{\sum_{i=1}^N (r_{\lambda}^{TOA}(\text{SNAP})_i - r_{\lambda}^{TOA}(\text{GEE})_i)^2}{N}} \quad (3)$$

$$\%NRMSE_{TOA} = \frac{RMSE_{TOA}}{[r_{\lambda}^{TOA}(\text{SNAP})_{\max} - r_{\lambda}^{TOA}(\text{SNAP})_{\min}]} \times 100 \quad (4)$$

$$RMSE_{BRR} = \sqrt{\frac{\sum_{i=1}^N (r_{\lambda}^{BRR}(\text{SNAP})_i - r_{\lambda}^{BRR}(\text{GEE})_i)^2}{N}} \quad (5)$$

$$\%NRMSE_{BRR} = \frac{RMSE_{BRR}}{[r_{\lambda}^{BRR}(\text{SNAP})_{\max} - r_{\lambda}^{BRR}(\text{SNAP})_{\min}]} \times 100 \quad (6)$$

Where: RMSE is root mean squared error; %NRMSE is percentage normalized root mean square error; $r_{\lambda}^{TOA}(\text{GEE})$ and $r_{\lambda}^{TOA}(\text{SNAP})$ are TOA reflectance derived using GEE and SNAP platform respectively for a particular wavelength (λ); $r_{\lambda}^{TOA}(\text{SNAP})_{\max}$ is maximum value of TOA reflectance derived within SNAP for a particular wavelength (λ); $r_{\lambda}^{TOA}(\text{SNAP})_{\min}$ is minimum value of TOA reflectance derived within SNAP for a particular wavelength (λ); $r_{\lambda}^{BRR}(\text{GEE})$ is Rayleigh corrected reflectance derived within GEE interface for a particular wavelength (λ); $r_{\lambda}^{BRR}(\text{SNAP})$ is Rayleigh corrected reflectance derived within SNAP for a particular wavelength (λ); $r_{\lambda}^{BRR}(\text{SNAP})_{\max}$ is maximum value of r_{λ}^{BRR} derived within SNAP for a particular wavelength (λ); $r_{\lambda}^{BRR}(\text{SNAP})_{\min}$ is minimum value of r_{λ}^{BRR} derived within SNAP for a particular wavelength (λ); N is the total number of samples (in this case pixels) used for error estimation.

2.4. CyanoKhoj Dashboard

CyanoKhoj GEE dashboard was developed for water quality monitoring using Sentinel 3-OLCI data dedicated to CyanoHABs, which features the 620 nm spectral band for accurate and near real-time detection of CyanoHABs at a revisit frequency ($\sim$ daily). CyanoKhoj provides many visualizations such as Normalized Difference Chlorophyll Index (NDCI) for Chl-a (Mishra and Mishra, 2012), CCD (Lunetta et al., 2015; Wynne et al., 2008) by incorporating previously established models (Eqs. (7), 8, 9, 10, 11, 12, and (13) as JavaScript code in GEE platform. Lunetta et al. (2015) CCD model (Eq. (13)), which builds on the Wynne et al. (2008) model (Eq. (9)) by decreasing the false positives caused by other types of

algae blooms, has been adopted by CyanoKhoj and is also adopted by National Oceanic and Atmospheric Administration (NOAA) for routine CCD mapping. CyanoKhoj further uses the Mishra and Mishra (2012) NDCI model to additionally delineate between other algal blooms and CyanoHABs. In addition to visualization, CyanoKhoj also facilitates analysis in terms of all the parameters. For example, when a pixel of the satellite image is clicked upon, the index value (e.g., NDCI or Cyano-bacteria Index (CI)) or concentration (e.g., Chl-a or CCD) is displayed for that pixel. Apart from this, clicking on any pixel of the satellite image generates a spectrum of the $r_{\lambda}^{BRR}(\lambda)$ across the wavelengths of the 21 bands of OLCI. These spectra can be analysed to detect an ongoing CyanoHAB or algae bloom or even the presence of cyanobacterial cells in the waterbody at a specific location pertaining to the pixel that was clicked upon.

$$NDCI = \frac{r_{\lambda}^{BRR}(709) - r_{\lambda}^{BRR}(665)}{r_{\lambda}^{BRR}(709) + r_{\lambda}^{BRR}(665)} \quad (7)$$

$$\text{Chl a} = 194.325 \times NDCI^2 + 86.115 \times NDCI + 14.039 \quad (8)$$

$$SS(681) = r_{\lambda}^{BRR}(681) - r_{\lambda}^{BRR}(665) - [(r_{\lambda}^{BRR}(709) - r_{\lambda}^{BRR}(665))] \times 0.3636 \quad (9)$$

$$CI = -1 \times SS(681) \quad (10)$$

$$SS(665) = r_{\lambda}^{BRR}(665) - r_{\lambda}^{BRR}(620) - [(r_{\lambda}^{BRR}(681) - r_{\lambda}^{BRR}(620))] \times 0.7377 \quad (11)$$

$$\text{If, } SS(665) > 0 \text{ then } CI = CI_{\text{cyano}} \quad (12)$$

$$CCD = CI_{\text{cyano}} \times 10^8 \quad (13)$$

Where:

NDCI=normalized difference chlorophyll index; Chl-a=Chlorophyll-a; SS (681) = Spectral Shape at 681 nm; SS (665) = Spectral Shape at 665 nm; CI = Cyanobacteria Index; CI_{cyano} = Cyanobacteria Index specific to cyanobacteria only; CCD = cyanobacteria cell density.

Although the CCD model presented above (Eq. (13)) was validated (MAPE = 28.6%) in a wide range of inland and coastal waters within the United States (Lunetta et al., 2015), we were able to validate this model for only one Indian site, Chilika Lagoon, Odisha, India due to the lack of in-situ data from other waterbodies. Two years of monthly sampled CCD data (2017–2018) from Chilika Lagoon (Data provided by Chilika Development Authority, India) was utilized during validation. We only used non-zero CCD data ($n = 29$) during validation and excluded samples near land to avoid any adjacency effect. Following the above criteria, CCD data were derived within GEE using Lunetta et al. (2015) CCD model (Eq. (13)) for matched-up samples. A linear regression analysis was carried out between in-situ and satellite derived CCD for matched-up samples, and %NRMSE was estimated.

3. Results and Discussion

CyanoKhoj framework uses widely established calibrated and validated models for visualization of CyanoHABs products. For example, the NDCI model for Chl-a had a mean standard error of 2.49 mg/m³ in its validation across numerous waterbodies and trophic status from hypereutrophic to oligotrophic waters (Mishra and Mishra, 2012). The CCD model (CI algorithm) originally developed by Wynne et al. (2008, 2010) and modified by Lunetta et al. (2015) has been validated in waterbodies from Florida, Ohio, and New England states, producing a mean absolute percentage error (MAPE) of 28.6% for a range between 10,000 to >1 million cells/ml (Clark et al., 2017). Another recent study in early 2021 by Mishra et al. (2021) also validated this algorithm for CyanoHAB detection by using field data from 30 lakes in the United States and reported an accuracy of 84% with $\sim$ 90% cyanobacteria positives correctly classified. Again, we validated Lunetta et al. (2015) model in Chilika Lagoon, India, with total 29 samples ($n = 29$), which

were found to match-up on the same-day cloud-free satellite overpass. A significant correlation ($R^2 = 0.39$; $p < 0.01$; $n = 29$) was observed between in-situ versus satellite derived CCD and %NRMSE was estimated 21.46%. Site-specific tuning of the CCD model would be required in the future to improve the accuracy when more in-situ data become available from Indian waterbodies. Nevertheless, the main goal of this study was not the model calibration-validation but to showcase the utility of the tool for rapid monitoring and discuss the widespread CyanoHABs problem across India.

We present the first country-wide CyanoHABs map for India generated using CyanoKhoj (Fig. 2). All inland and coastal waters, which are resolved by the 300-m OLCI sensor, are presented in the form of cloud-free time-series composite maps. The CCD composite for India was prepared by selecting all available images during September-October 2018 and using various features of CyanoKhoj such as cloud masking, non-water pixel masking, implementing CCD algorithm, and generating maps. Before implementing the CCD algorithm to the composite image, reflectance values of cloud-free pixels were averaged for images

between two months (September-October). September-October time window was selected to showcase the utility of the tool because that is one of the time periods when algae bloom in Indian waters peak coinciding with the withdrawal of the southwest monsoon (D'Silva et al., 2012; Mishra et al., 2020b). Although we highlight its utility for Indian waters, CyanoKhoj is not limited to a specific time period or region. It is developed to map global CyanoHABs occurrences using all archived and future Sentinel-3A/B OLCI data. In addition, the tool can be implemented on the now decommissioned ESA's Medium Resolution Imaging Spectrometer (MERIS) satellite archived data from 2002 to 2012, which had similar band-architecture and spatial resolution as Sentinel-3. The prospect of mapping and analyzing the extent, frequency, and magnitude of CyanoHABs globally or specific to a region on a semi-continuous long-term scale of almost two decades using MERIS and Sentinel-3 makes this a powerful tool.

CyanoHABs are a result of climate change induced warming and nutrient pollutions (Mishra et al., 2020a; O'Keefe and Juliette 2019) and are driven by many factors such as land surface runoff, wastewater

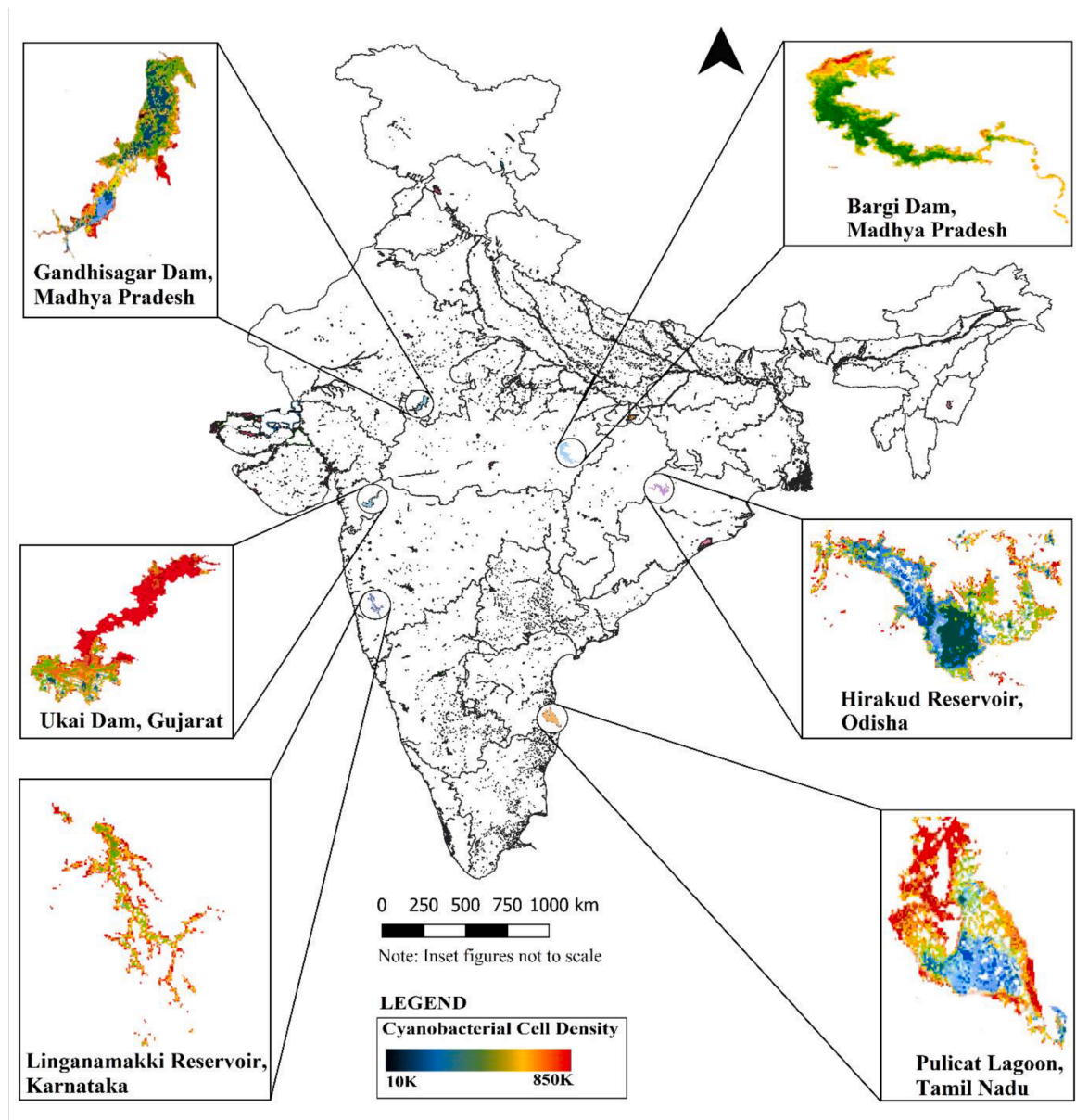


Fig. 2. CyanoHABs distribution in Indian waters in September-October 2018 was generated by CyanoKhoj. Inset: Six sites from different parts of India are highlighted as examples.

discharge, seasonal variability, cyclones, and varying monsoons (Lévy et al., 2007; Srichandan et al., 2015; Vinayachandran and Mathew, 2003). Although tens of thousands of waterbodies were resolved and mapped by this tool, we selected and analysed six sites from the CyanoKhoj map (Fig. 2) representing different geographic and climatic regions, from arid western India (Ukai Dam, Gujarat), to warm central India (Bargi and Gandhisagar Dams in Madhya Pradesh), and from hot and humid coastal states in west, south, and east India (Linganamakki Reservoir, Karnataka; Pulicat Lagoon, Tamil Nadu; and Hirakud Reservoir, Odisha). In 2018, heavy algal blooms were reported at these sites by Mishra et al. (2020a). CyanoHABs for many locations in India are often driven by rainfall variability and dam-regulated discharge (Acharyya et al., 2012). Waterflow at reservoir sites is primarily controlled by dams. CyanoKhoj generated spatio-temporal maps for these sites showed a noticeable temporal change in Chl-a and CCD levels. A clear temporal evolution of the CyanoHABs can be visualized for Bargi Dam (Fig. 3), Gandhisagar Dam (Fig. 4), Ukai Dam (Fig. 5), Linganamakki Reservoir (Fig. 6), Pulicat Lagoon (Fig. 7), and Hirakud Reservoir (Fig. 8) during Sept-Nov 2018.

The three western and central reservoirs (Bargi, Gandhisagar, and Ukai) were eutrophic to hyper-eutrophic from September to November, but the initiation and growth of CyanoHABs were mostly restricted temporally and to specific spatial locations. Bloom phenology can be seen for all three sites in the form of early-onset, full-on CyanoHABs, and a gradual decline. In the case of Bargi Dam (Fig. 3), the CCD growth can be visualized from the early onset in mid-September, which spreads across the lake by the end of October and eventually starts to decline in late November. Indian monsoon typically begins in June and lasts till September, while October to December is the post-monsoon period in which the monsoon retreats. Rasal et al. (2019) showed that the phosphate content level post-monsoon (starting September) was the highest compared to the other three seasons (Summer, Monsoon, and Winter) in the Bargi Dam, which was attributed as a major factor for the bloom. In the case of Gandhisagar Dam (Fig. 4) and Ukai Dam (Fig. 5), a more intense concentration and spread of the CyanoHABs can be seen towards the end of September and early October, and a gradual decline as November progresses. *Microcystis* was found as the dominant cyanobacteria species in Gandhisagar Dam in earlier studies (Kumari et al., 2018; Rao and Choubey, 1990). The CyanoHAB maps for Linganamakki Reservoir, the largest reservoir in Karnataka damming the Sharavathi River, showed CCD peaking towards the end of the year from

mid-October to early December (Fig. 6). The nutrient content in the reservoir is typically at the highest level in the post-monsoon period, which drives productivity in the late October-December months. Although past studies have found it to be oligotrophic with an average net production of $125 \text{ mg C/m}^2/\text{h}$, *Microcystis* has been found in the reservoir, along with 45 mg/100 g of soil of essential nutrients and a high bacterial activity of 27 C/N ratio (Ramakrishniah et al., 2000). Moreover, a 2008 study of Sharavati Basin also hinted at high phytoplankton diversity in the Linganamakki Reservoir (Desai et al., 2008). Pulicat Lagoon, the second largest lagoon in India and located on the southeast coast of Tamil Nadu, showed severe CyanoHABs during Sep-Nov 2018 (Fig. 7). The lagoon is rich in nutrients and diatoms with dense seagrass cultivation, and a study on the diatom blooms conducted in 2015, showed a rise in the physio-chemical characteristics of the lagoon (Santhanam et al., 2018). Being a fragile ecosystem, it has become increasingly prone to CyanoHABs due to the high nutrient content, temperature, and irregular freshwater discharge. The Hirakud Reservoir, the world's largest earthen dam in western Odisha, showed the onset and decline of a CyanoHAB from late September to early November in 2018 although the spatial distribution and CCD range was comparative smaller to other waterbodies in this study (Fig. 8). This reservoir, which dams the Mahanadi River, was built primarily for electricity generation but is now additionally being utilized for fisheries and drinking water. It has been directly polluted by the untreated discharge of manufacturing plants and industries in the district and has become prone to toxic CyanoHABs and increased dominance to other cyanobacteria species such as *Anabaena*, *Oscillatoria*, *Chroococcus*, and *Phormidium* (Deep et al., 2013; Ramya et al., 2020). A study of the Mahanadi River in 2015 showed the presence of 37 cyanobacteria species, out of which 7 were recorded as toxic (Niemen-Ebrott et al., 2015).

All the above waterbodies shown as case studies are significantly large and provide important ecological and economic services to the states and surrounding populations. However, they are practically data-scarce regions as we did not find any synoptic monitoring studies for these sites. There are only a handful of studies which inspect the water quality of these sites by in situ analysis, none of which are CyanoHAB exclusive. Their large size make them suitable for satellite-based mapping, which can be a great tool to monitor these waterbodies continuously that simply cannot be achieved by in situ samplings. A vast number of people directly depend on these waterbodies for their

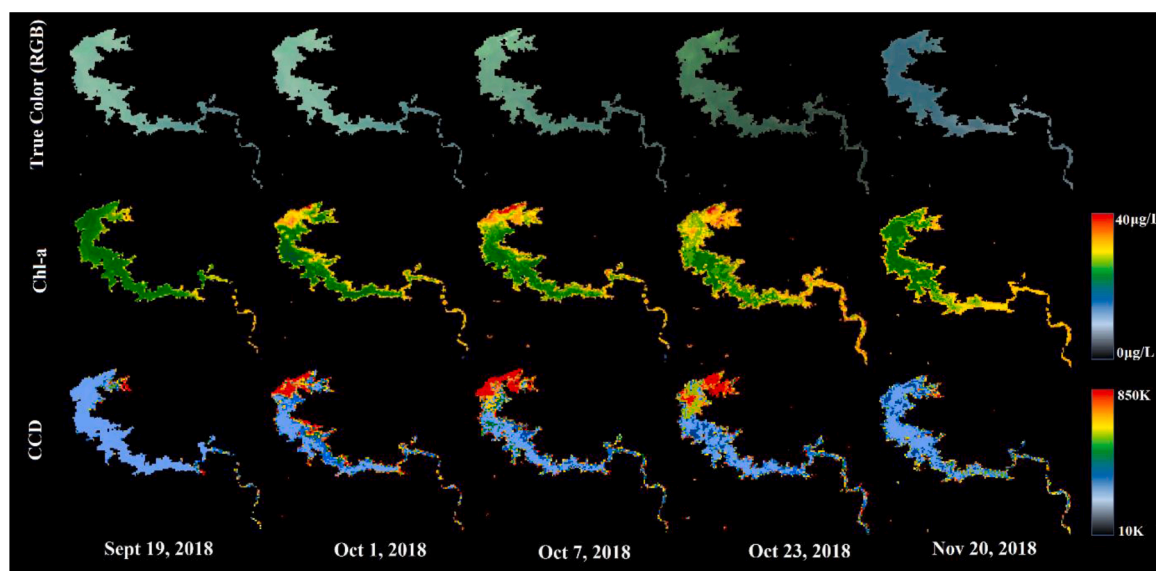


Fig. 3. The spatio-temporal evolution of CyanoHABs at Bargi Dam (Madhya Pradesh). Row 1 shows the Sentinel-3 True Color (R, G, B: Bands 8, 6, 4) product, Row 2 shows the CyanoKhoj generated Chl-a concentration maps, and Row 3 shows the CCD maps.

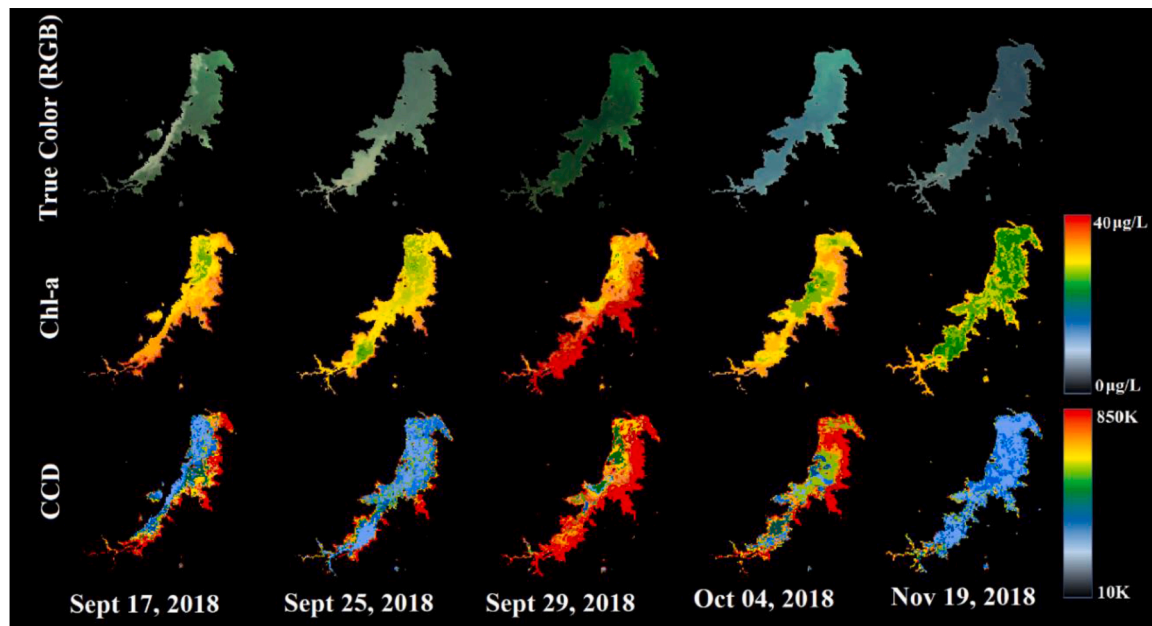


Fig. 4. The spatio-temporal evolution of CyanoHABs at Gandhisagar Dam (Madhya Pradesh). Row 1 shows the Sentinel-3 True Color (R, G, B: Bands 8, 6, 4) product, Row 2 shows the CyanoKhoj generated Chl-a concentration maps, and Row 3 shows the CCD maps.

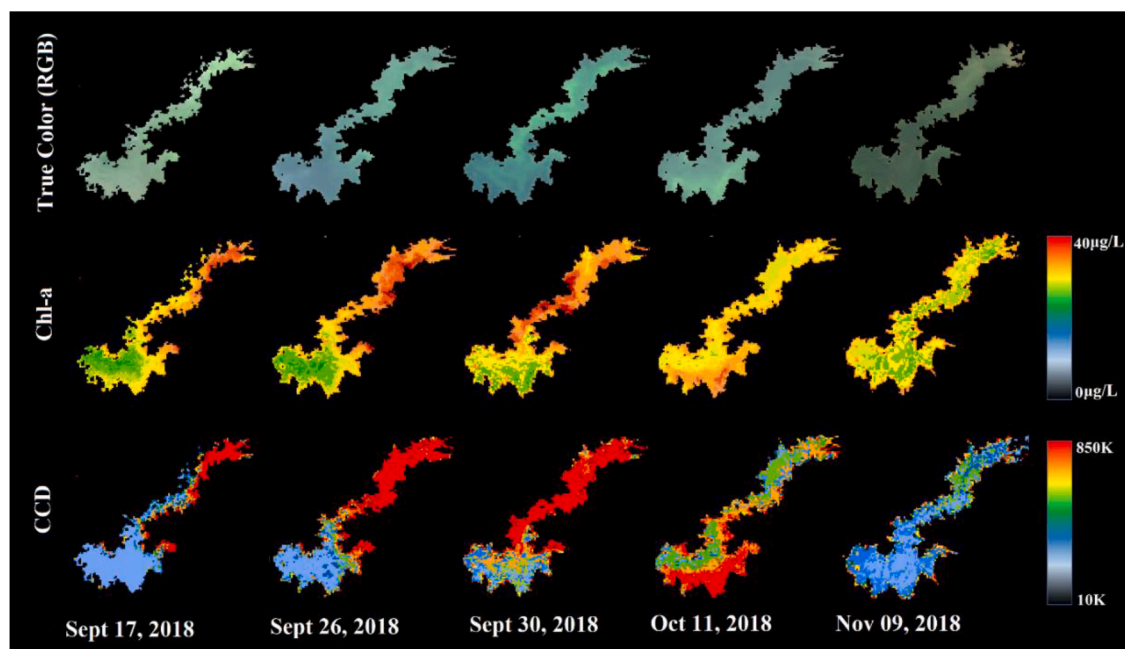


Fig. 5. The spatio-temporal evolution of CyanoHABs at Ukai Dam (Gujarat). Row 1 shows the Sentinel-3 True Color (R, G, B: Bands 8, 6, 4) product, Row 2 shows the CyanoKhoj generated Chl-a concentration maps, and Row 3 shows the CCD maps.

livelihoods, and a lack of CyanoHAB monitoring protocol makes them exposed and vulnerable to the toxin impact. If these large and vitally important waterbodies are not monitored frequently, then the chance of smaller and less important waterbodies being monitored is negligible. This provides the rationale for our study, especially because blue-green algae (cyanobacteria) is mostly the dominant phytoplankton community in the man-made reservoirs (Kumari et al., 2018). South Africa would be the best example where most of the man-made reservoirs have a CyanoHAB problem (Matthews, 2014). However, the South African government has funded CyanoLakes (CyanoLakes, 2015) to monitor its reservoirs using satellite data. CyanoKhoj can emerge as a monitoring tool not only for Indian waterbodies but for global waterbodies. In the

United States, various state agencies are looking for satellite-based CyanoHABs monitoring options. For example, now California has its own satellite-based CyanoHABs website supported by NOAA (<https://flab.sfei.org/>), Minnesota has its own LakeBrowser that monitors important water quality indicators (<https://lakes.rs.umn.edu/>). NOAA is also monitoring six large waterbodies across six different states in U.S. (Florida, Louisiana, Michigan, Wisconsin, North Carolina, and Ohio) using satellite data in their new web portal (NOAA, 2019), apart from employing a citizen science program to report harmful algal blooms in the state of Florida (Ransom Hardison et al., 2019). CyanoKhoj is different and unique from the above monitoring systems in terms of user's control over input and output products. It is a GEE-based

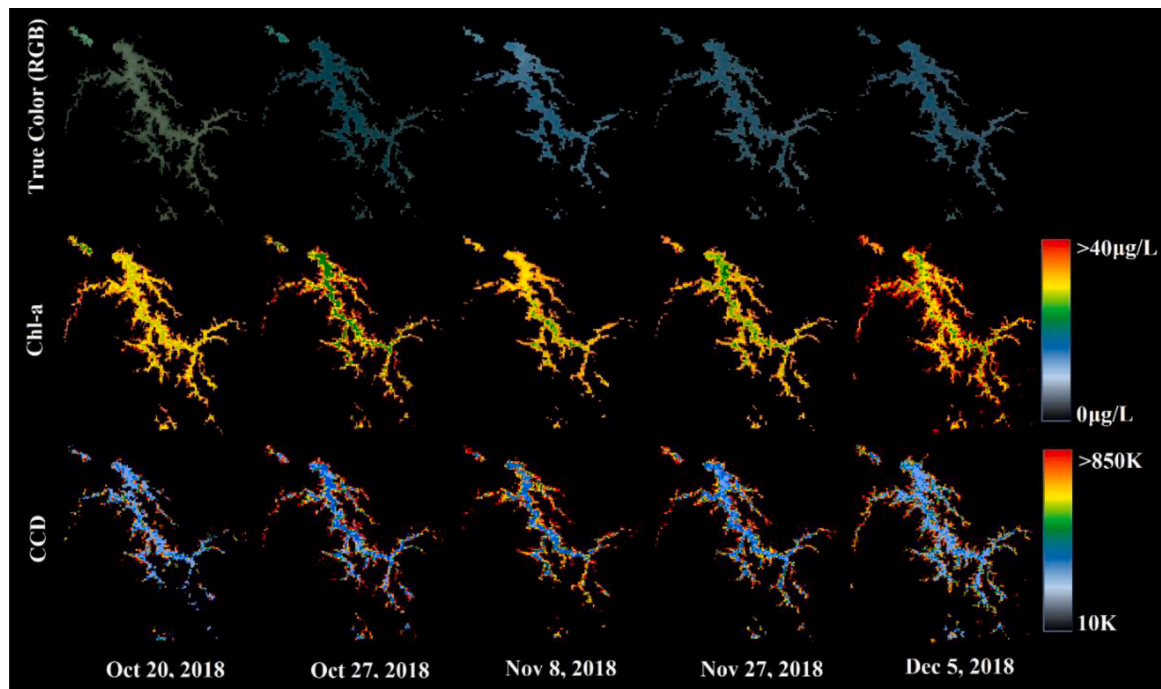


Fig. 6. The spatio-temporal evolution of CyanoHABs at Linganamakki Reservoir (Karnataka). Row 1 shows the Sentinel-3 True Color (R, G, B: Bands 8, 6, 4) product, Row 2 shows the CyanoKhoj generated Chl-a concentration maps, and Row 3 shows the CCD maps.

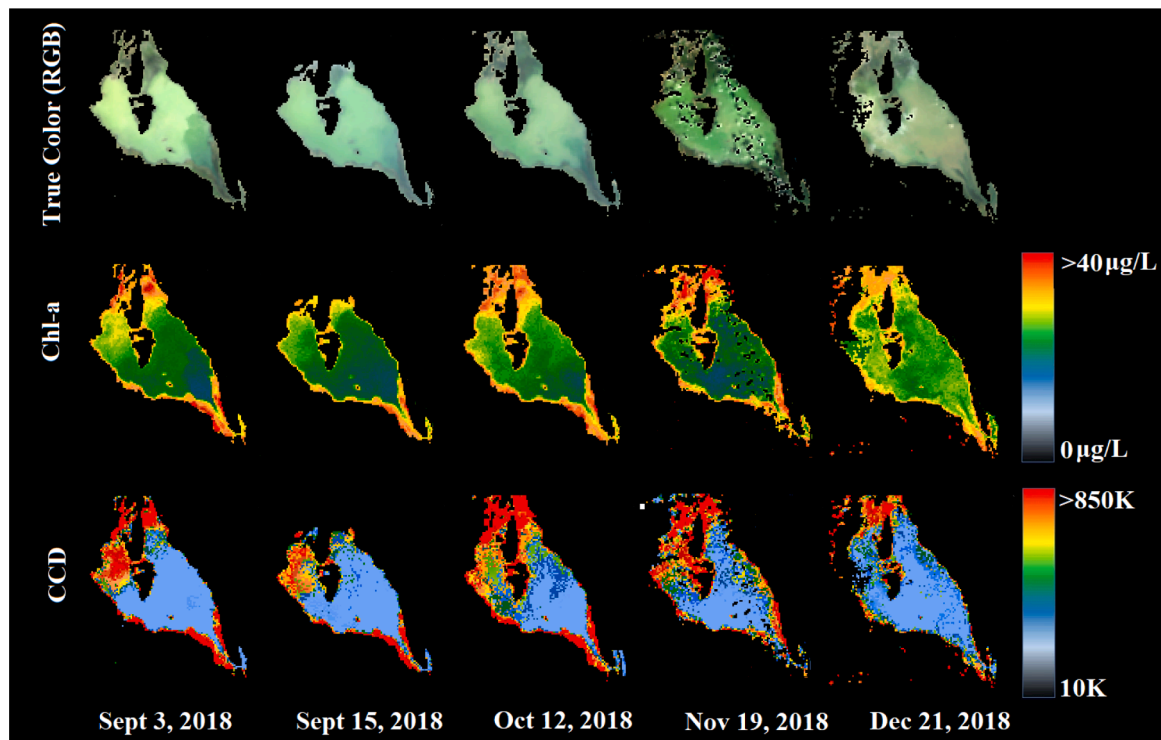


Fig. 7. The spatio-temporal evolution of CyanoHABs at Pulicat Lake (Tamil Nadu). Row 1 shows the Sentinel-3 True Color (R, G, B: Bands 8, 6, 4) product, Row 2 shows the CyanoKhoj generated Chl-a concentration maps, and Row 3 shows the CCD maps.

interactive web-app that allows quick and near-real-time analysis of waterbodies globally for CyanoHABs. Users can navigate to any waterbody worldwide, which can be resolved in 300 m Sentinel-3 OLCI imagery, for detection and mapping CyanoHABs. Apart from providing the Chl-a and CCD maps of the waterbody, CyanoKhoj has the unique feature of analysing the spectra of any pixel on the image to further zero

down on the CyanoHAB analysis. Moreover, the GEE code for CyanoKhoj will be available as open access that will allow users to tweak the code using site-specific cyanobacteria data to improve model prediction. Also, users do not have to pay for any subscription such as other commercial monitoring platform “CyanoLakes”. [Section 2](#) of the supplementary section highlights the user-interaction as well as the unique

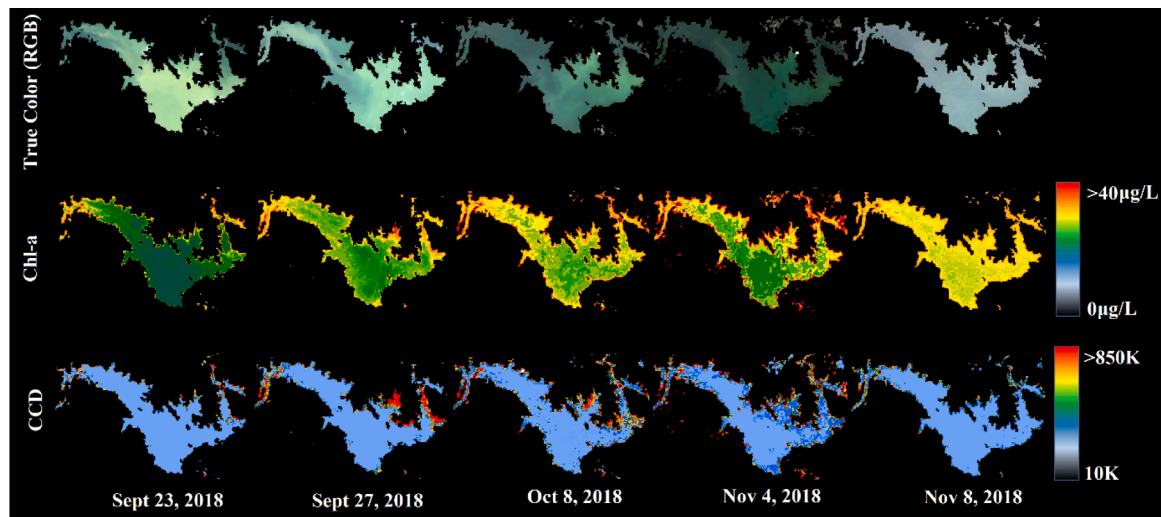


Fig. 8. The spatio-temporal evolution of CyanoHABs at Hirakud Reservoir (Odisha). Row 1 shows the Sentinel-3 True Color (R, G, B: Bands 8, 6, 4) product, Row 2 shows the CyanoKhoj generated Chl-a concentration maps, and Row 3 shows the CCD maps.

features of CyanoKhoj.

Further, to observe the inter-annual variability of CyanoHABs, we extracted monthly averaged CCD data from an identified spatial “hot-spot” for each study site and produced a four-year time series for the above six study sites (Fig. 9). Monthly aggregation was performed by creating a median composite scene out of all cloud-free scenes available in that month, and the mean CCD value was calculated for each study site’s hotspot. There are occasional discontinuities in the time series owing to inevitable data gaps due to cloud cover (Fig. 9). Though the severity of CyanoHABs may vary throughout the lakes, each site showed a consistent seasonal phenology in the evolution and decadence of cyanobacteria. Ukai Lake, Barghi Reservoir and Gandhisagar Dam, which are inland waters, become most susceptible to intense CyanoHABs during the post-monsoon season, and the blooms finally subside by late winter (Fig. 9). Hirakud Reservoir, another inland waterbody, also showed a similar trend, but the blooms persisted for a longer time and finally dissipated during the early summer months. Lignanmakki, a marshy inland water site, was found to be suffering from heavy CyanoHABs for longer period throughout the year, roughly spanning from early monsoon to late summer with CCD peaks during winter. Pulicat Lake, which is not entirely inland due to the presence of coastal transitions, undergoes CyanoHABs consistently throughout the year, with long peaks in late summer/ early monsoon and almost no months bloom-

free.

3.1. Potential Broader Impact of CyanoKhoj on Indian Freshwater Management

A cloud-based rapid freshwater monitoring tool is much needed in India, where water pollution continues to be one of the most prominent environmental problems exacerbated in the past decade owing to rapid urbanization and industrialization (Garg, 2012; Williams et al., 2018). The major cause of the dire condition of waterbodies in India is the improper discharge of waste materials and their management methods (Kumar et al., 2017). Around 80% of urban waste discharge ends up in rivers and reservoirs globally (UN WWDR, 2017). The pollutants, along with nutrients, then spread indirectly to all the areas in the country and, coupled with climate change-induced warming, result in severe eutrophication in waterbodies. Many studies have reported heavy algal blooms and eutrophication in various lakes and ponds and the coastal regions throughout India (Al Shehhi et al., 2014; Bhavya et al., 2016; Kundu et al., 2015; Ray et al., 2020; Yannawar and Bhosle, 2013). However, very few studies differentiate between phytoplankton and toxic CyanoHABs and examine the cause of extreme adversities of waterborne diseases, which affect not only the citizens’ health and life but also the aquatic ecosystem (Kumar et al., 2017). The numbers are

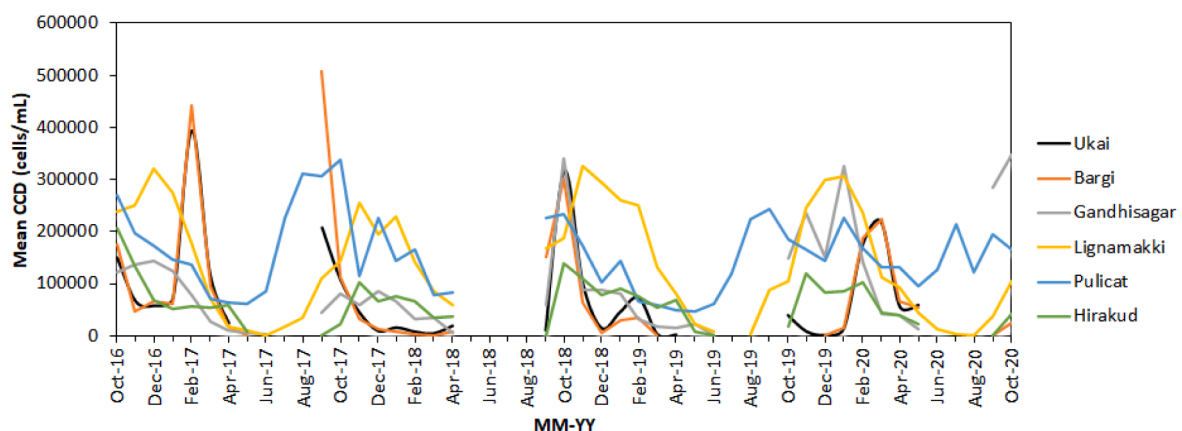


Fig. 9. A four-year monthly time series of cyanobacteria cell density (CCD) for all study sites (Ukai Dam, Bargi Dam, Gandhisagar Dam, Lignanmakki Reservoir, Pulicat Lake, and Hirakud Reservoir) showing a strong seasonality in cyanobacteria blooms.

hard to fathom. For example, roughly 2500 people died, and more than 13 million were diagnosed with water-related diseases in 2018 alone (MoHFW, 2018; TimesOfIndia, 2018). Degraded water quality not only contributes to water scarcity but also dents the economy of the country severely (Murty and Kumar, 2011). The central and state governments of India have been implementing measures to handle water pollution for a long time. Two of the most important examples include the *Water Prevention and Control of Pollution Act, 1974* and the *Water Prevention and Control of Pollution Cess Act, 2003* (IndiaCode, 2013), which prohibit harmful discharge in waterbodies and promote the prevention of water contamination. The Central Pollution Control Board (CPCB) of India has the prime responsibility of the wellness of Indian waterbodies. CPCB has, for this motive, formed a National Water Quality Monitoring Network, which monitors the water quality of all Indian waterbodies by routinely analysing water samples for water quality parameters except CyanoHABs (CPCB, 2015a). In 2014, CPCB also mandated the policy of Zero Liquid Discharge (ZLD) at the state level, which mandates that industries must reuse the industrial and municipal wastewater instead of discharging it into waterbodies (CPCB, 2015b). Apart from this, various river-specific (Ganges, Yamuna, Cocum, Adyar) cleaning drives have been carried out jointly by CPCB and other independent organizations. In 2018, NITI Aayog, a think tank of the Indian government, developed a Composite Water Management Index (CWMI), a novel metric to measure water management throughout the country by various data collection and water analysis activities (CWMI-NITI Aayog, 2019).

While these infrastructure(s) and implementations discussed in the previous section have been carried out country-wide, chronological analyses of various water quality audit reports of the last decade indicate a lot of pressing issues. Performance audits for water quality in 2011 summarized that the water pollution issue had not been addressed adequately in terms of effective implementation (MoEF, 2011). For example, there was an inadequate number of monitoring stations for tracking pollutions in rivers and lakes, a lack of real-time monitoring of water quality, and data on water quality had not been disseminated properly. The CPCB 2015 report stressed freshwater being a vulnerable resource and recommended various awareness programs for water quality while addressing the inadequacies in water pollution prevention and monitoring guidelines (MoEF, 2015). The CWMI report of 2019 noted that the overall performance of the country in tackling water challenges has been poor and stressed the proper implementation of water quality monitoring and management solutions. In summary, India's inland water quality is poor, and people's exposure to these degraded waters is high. That makes CyanoKhoj as a monitoring and decision-making tool vital from a public health perspective because ingesting toxic-cyanobacteria is a real threat to Indian people, and CyanoKhoj can help identify waterbodies and delineate areas within the waterbodies where CyanoHABs is a threat so that timely warnings can be issued.

4. Conclusion

The current scenario in India makes it imperative to have a country-wide water quality monitoring infrastructure, which can go hand-in-hand with the government policies for the same. As discussed earlier, pervasive eutrophication and CyanoHABs in Indian inland waterbodies still remain a pressing issue given their implications. This study showcases a first-ever all-India CCD map for the year 2018, to visualize the spatial distribution of CyanoHABs throughout the country. Furthermore, a cloud-based geospatial platform, called CyanoKhoj, is also proposed as a means for quick detection of CyanoHABs and near-real-time monitoring of inland and coastal water bodies. CyanoKhoj is an interactive, user-friendly tool that employs satellite image processing to monitor waterbodies specifically for CyanoHABs and surface water pollution. It is hosted online as a GEE web-app and is available for use (CyanoKhoj, 2021). It can readily be used by water managers and Non-Governmental Agencies (NGOs) to monitor Indian waterbodies for water pollution,

specifically for CyanoHABs and ordinary algal blooms, as an early warning system. This can help in maintaining water quality around the country and reducing waterborne health hazards and improving the aquatic ecosystems. CyanoKhoj is easy to use and share, which may encourage water resources managers to include it in their monitoring protocol to effectively target, detect, and visualize the bloom-affected waterbodies. It will also be helpful for planning field trips to such sites and executing timely management strategies such as targeted sampling. Apart from CyanoHABs monitoring, CyanoKhoj also has a lot of on-the-fly interpretation and visualization features such as on-click reflectance spectra for any pixel of the image, on-click values of various parameters such as NDCl, Chl-a, CCD, various masks to remove clouds and non-water pixels. Moreover, this dashboard is a completely cloud-based platform, so only a browser and an active internet connection are needed to use it. This makes it computationally inexpensive and increases the availability and accessibility of the dashboard to not only resource managers but also concerned citizen scientists and the laymen community.

In the future, testing and finetuning the models' performance in an intensive validation campaign across a broad range of Indian waterbodies would be necessary. While the current CCD model performed well in inland and freshwater bodies, it has a significant scope for improvement in coastal lagoons and brackish waterbodies. It is also extremely important to have extensive field data to generally improve the quantification across inland and coastal waterbodies in India, given the varied spectra of aquatic ecosystems that thrive in the country. This leads to a need for a large, temporally updated, and accessible CCD field database to further calibrate such models. Additional challenges in terms of image processing issues such as benthic interference in shallow or clear waters, and refinement of the non-water pixel mask should also be addressed. CyanoKhoj currently uses optical data of 300 m spatial resolution from the Sentinel-3 OLCI sensor for CyanoHAB monitoring. This limits the analyses to larger waterbodies, and smaller lakes and ponds are difficult to analyze. Since CyanoHABs are directly linked with temperature and nutrients, CyanoKhoj can be made more robust by integrating data from multiple sensors such as Landsat or MODIS derived temperature products, and also high-resolution Sentinel 2 data for a cross-calibrated model, which can be more robust in terms of the detection and quantification of CCD and PC by combining data from climatically varied geographical sites (Page et al., 2018).

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.hal.2021.102160.

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